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Creating a Private Subnet



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The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation at the top indicates the path: VPC > Subnets > Create subnet. The page title is 'Subnet settings' with a subtitle 'Specify the CIDR blocks and Availability Zone for the subnet.'

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Text input field:
The name can be up to 256 characters long.

Availability Zone info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Dropdown menu:

IPv4 VPC CIDR block info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
Dropdown menu:

IPv4 subnet CIDR block
Text input field: 256 IPs

Tags - optional

Key	Value - optional	
Name	NextWork Private Subnet	Remove

You can add 49 more tags.

The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a virtual network in AWS where you can launch and manage resources like EC2 instances in an isolated environment. It's useful because it gives you full control over network configuration, including subnets, route tables, security groups, and network ACLs, allowing you to secure resources, manage traffic, and separate public and private workloads.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a private subnet, attach a dedicated route table, and set up a custom network ACL to keep resources isolated and secure from the internet.

One thing I didn't expect in this project was...

One thing I didn't expect was that a custom network ACL starts by denying all traffic, even within the VPC, requiring explicit rules later to allow communication.



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This project took me...

This project took me about an hour.

Private vs Public Subnets

The difference between the both is that a public subnet has a route to the internet through an internet gateway, while a private subnet has no direct internet route, keeping its resources isolated.

Private subnets exist to keep sensitive resources like databases and internal services isolated from the internet, adding security and control within a VPC.

My private and public subnets can't share the same route table, because their traffic rules are different . public subnets route to the internet, private subnets do not.

The screenshot displays the AWS Management Console interface for creating a new subnet. The breadcrumb navigation shows 'VPC > Subnets > Create subnet'. The page title is 'Subnet settings' with a subtitle 'Specify the CIDR blocks and Availability Zone for the subnet.' Below this, it indicates 'Subnet 1 of 1'. The 'Subnet name' field is 'NextWork Private Subnet'. The 'Availability Zone' is set to 'Middle East (Bahrain) / me1-south-1a'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.1.0/24'. At the bottom, there is a 'Tags - optional' section with a key 'Name' and a value 'NextWork Private Subnet'.

A dedicated route table

By default, my private subnet is associated with the main route table that AWS automatically creates for the VPC.

I set up a new route table to give the private subnet its own traffic rules, keeping it separate from the public subnet and blocking direct internet access.

The private subnet's route table allows traffic within the VPC, such as communication between subnets, but does not route traffic directly to the internet.

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Updated

Endpoints

Updated

Endpoint services

Route tables (1/3)

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒ NextWork Public Route Table	rtb-04478d859bd3365f4	subnet-0bef7d9136f745...	-	Yes	vpc-061036d8
☐ -	rtb-053e51e561c573928	-	-	Yes	vpc-0f93cd9e
☐ NextWork Private Route Table	rtb-03d87a8b77b76d91f	subnet-0f8fab4da02da3...	-	No	vpc-061036d8

rtb-04478d859bd3365f4 / NextWork Public Route Table

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Route table ID

rtb-04478d859bd3365f4

Main

Yes

Explicit subnet associations

subnet-0bef7d9136f745b74 / NextWork Public Subnet

Owner ID

vpc-061036d8161543daf | NextWork VPC

881490105972

Edge associations

-

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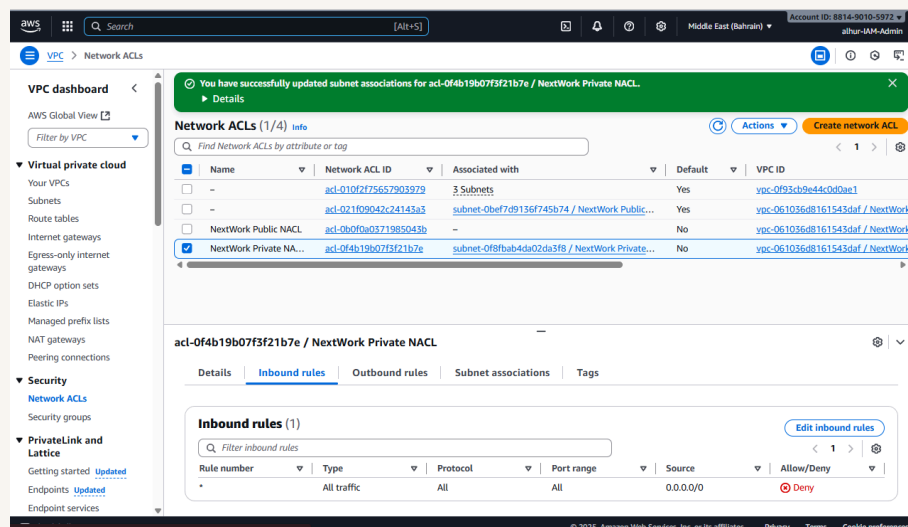


A new network ACL

By default, the private subnet is associated with the VPC's default network ACL, which allows all inbound and outbound traffic.

i set up a new network ACL to customize and control traffic for the private subnet, adding extra security beyond the default ACL.

The inbound and outbound rules in the new private NACL are both set to deny all traffic by default. Custom NACLs start with all traffic blocked to secure the subnet, and rules can be added later to allow specific traffic.





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